

Geothermal System Modelling

GEOptimize will use the building energy models to determine the impact in the size and cost of the GHX by performing a sensitivity analysis using Ground Loop Design (GLD 2016) software. GEOptimize will quantify the amount of additional kBtu's needed to be extracted from or rejected to the GHX to obtain an annual energy balance to minimize the size of the GHX. The energy sources/sinks that will be investigated may include: horizontal ground heat exchangers, lake heat exchangers, solar thermal panels, and vertical ground heat exchangers. A series of simulations will be developed to optimize system performance including:

- The size and configuration of GHX modules,
- Location of GHX modules in relation to building heating and cooling requirements,
- System control strategy,
- Additional waste and renewable energy resources and heat sinks and their impact on the size and performance of the GHX modules,

Scope of work

Phase 1 – Schematic Design

- Detailed energy modelling analysis for each building.
- Analysis of GHX requirements and auxiliary heating and cooling system.
- Verification of most appropriate heat exchanger solution.
- Layout of GHX circuit piping for the recommended GHX configuration.
- High level pricing.
- High level economic calculations including simple payback compared to a conventional system.

Phase 2 – Detailed Design Assistance

- Refine GHX & building energy models.
- AutoCAD drawings for GHX's up to manifold.
 - Design of fusion welded headers to connect GHX circuits to supply / return runout piping.
 - Supply / return runout piping from GHX headers to manifold in each tenant building.
 - Details of manifold.
- Heat pump and circulation pump sizing assistance.
- Coordination and training focused on GSHP system design.
- Conceptual control strategy including long-term monitoring recommendations.

Deliverables

Deliverables for this design assistance include:

- Detailed energy models in Trane Trace 3D+ or 700 and 8760 energy loads.
- Preliminary GHX design for the building site based on the land area available in a detailed PowerPoint presentation.
- AutoCAD drawings of GHX design.
- Assistance in specification package writing.

Pricing

Description	Estimated Time (Weeks)	Line Total
Schematic Design	6-7	\$67,500.00
Detailed Design Assistance	5-6	\$63,000.00
Total (CAD)	11 – 13	\$130,500.00

Terms

- Additional work out of scope will be charged at \$250/hr.
- Invoices will be submitted monthly for work completed to date.
- Payment is expected within 30 days of invoice.
- Site visits are charged at cost +5%.
- Fees include 10 one-hour virtual meetings. Additional meetings will be invoiced at \$250 per hour.

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